Report 1: Are Headtracking Measures an Indicator of Depressive Symptoms?

360° Videos VR Experiment — Preliminary Analysis

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Github link for code: https://github.com/bharath-gajawada/Data_Analysis_Services

1. Brief Introduction

This study investigates whether **head-movement patterns during 360° VR video exploration** can serve as objective indicators of depressive symptoms. Rather than relying solely on self-reported emotions and standardized questionnaires, we capture psychomotor behavior via headtracking — measuring how actively participants scan their virtual environment. The core idea: *depression may reduce exploratory head movement, providing an objective, behavioral marker*.

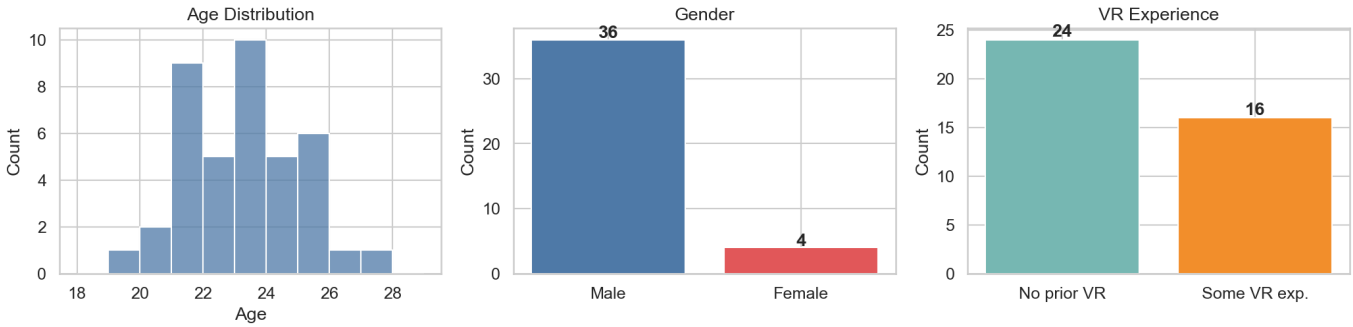
This is based on the work of Srivastava & Lahane (2025), who found that individuals with moderate-to-severe depression exhibited significantly slower scanning speeds during 360° video viewing, even after controlling for anxiety.

> Only self-reports alone miss important behavioral differences. So headtracking in VR may provide objective psychomotor markers for detecting depression.

2. Experiment Details

Component	Detail
Participants	N = 40 college students (36 male, 4 female), ages 19–27 (M = 22.78, SD = 1.80)
VR Headset	Meta Quest 3
Videos	5 × 360° videos spanning different emotional conditions
Surveys	PHQ-9, GAD-7, STAI-T, PANAS (pre & post VR), valence/arousal per video, Presence
Headtracking	Rotation data (Pitch/X, Yaw/Y, Roll/Z) + speed, recorded per video per participant

Figure 1: Participant Demographics (N = 40)



> We can see that we have significantly more male than female participants, college aged sample with limited prior VR experience, which is consistent with the reference study's population.

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3. Motivation for Study

Traditionally, methods of screening for depression use **self questionnaires** that are subjective and non objective. They may also not measure psychomotor behavior in real time. VR offers a medium through which it is possible to create a virtual environment where psychomotor behaviors can be measured objectively. In the reference paper (srivastava et al., 2025), it was established that **head movement scanning speed** when watching 360-degree videos was significantly low in people with moderate and severe depression. There were no differences in emotional rating between people with and without moderate and severe depression.

In this study, we investigate whether the same psychomotor behaviors can be seen in another group of people, whether head tracking can be used to differentiate between different groups of people with different levels of depression, and whether people with depression and anxiety have different behaviors compared to people with depression only.

> VR-based psychomotor assessment fills a gap in current methods, as objective behavioral responses are evident even when self reporting fails to distinguish between depressed and non-depressed people.

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4. Literature Review

Depression and Psychomotor Retardation. Clinical depression is associated with psychomotor disturbances, slowed physical movements and reactions (Choi et al., 2020). Since depression and anxiety frequently co-occur, studies examining behavioral markers must account for this comorbidity, as anxiety may independently influence psychomotor behavior.

Assessment Instruments. Depression severity is measured using the **PHQ-9** (Kroenke et al., 2001), anxiety using the **GAD-7** (Spitzer et al., 2006), and current mood using the **PANAS** (Watson et al., 1988). Emotional ratings (valence and arousal) follow Russell's (1980) Circumplex Model of Affect. VR presence is assessed using the Presence Questionnaire (Witmer & Singer, 1998).

VR as an Assessment Tool. Head-mounted VR provides controlled, immersive environments where 360° videos allow participants to freely explore, making head movement a natural proxy for psychomotor engagement.

Key Reference — Srivastava & Lahane (2025). In a study with N = 50 participants:

- Individuals with **moderate-to-severe depression showed significantly slower scanning speed** (M = 3.88 °/s, SD = 0.88) compared to mild (M = 4.94) and no depression (M = 5.29), with **large effect sizes** (d = −1.22 to −1.64).
- This effect held after controlling for anxiety via ANCOVA ($p < .001$, $\eta^2 = .295$).
- **Self-reported valence and arousal showed NO group differences** only the objective head movement data differentiated the groups.
- The **PHQ-9 × GAD-7 covariance** was strong (r = .752), motivating a "distress grouping" approach.

- Depression blunted responses to both positive and negative stimuli, with head movement uniformly reduced across all video types.

> Srivastava & Lahane (2025) demonstrated that head scanning speed in VR is a valid psychomotor marker, and our study attempts to replicate these findings with a new sample.

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5. Measures : What They Capture

PHQ-9 (Patient Health Questionnaire-9)

This is a list of nine questions in which users has to answer. Each question has a scale of 0 to 3 ("Not at all" to "Nearly every day"). in total the score ranges from 0 to 27. The categories are as follows according to the PHQ-9 manual:

Category	Score Range
Minimal	0–4
Mild	5–9
Moderate	10–14
Moderately Severe	15–19
Severe	20–27

GAD-7 (Generalized Anxiety Disorder-7)

This is a list of seven questions in which users has ansewer based on their experience in past two weeks.This is for measuring anxity in past two weeks. Each questions has 0-3 range totaling the range 0-21. The categories are as follows according to the GAD-7 manual:

Category	Score Range
Minimal	0–4
Mild	5–9
Moderate	10–14
Severe	15–21

STAI-T (State-Trait Anxiety Inventory — Trait)

This a list of 20 questions in which users has to answer. This is used for measuring long term anxity. This total score range is 20-80.

PANAS (Positive and Negative Affect Schedule)

This is a list of 20 questions in which users has to answer. This is used for measuring current mood. This is split into two parts Positive Affect (PA) and Negative Affect (NA). Each part has 10 questions. This is

administered before and after VR exposure to detect mood changes. The score range per subscale is 10-50.

Headtracking Measures

- **Scanning speed**: Rate of angular head movement (°/s) (degrees per second) — computed as total angular distance / time. Higher speed = more active exploration.
- **Yaw (Y-axis)**: Horizontal head turning, which is the primary scanning motion in 360° videos.
- **Pitch (X-axis)**: Vertical nodding.
- **Roll (Z-axis)**: Head tilting (minimal in VR viewing).
- **Standard deviation**: Variability of head position around the mean.

> PHQ-9 and GAD-7 use established clinical cutoffs for grouping; headtracking speed (especially yaw) is the primary behavioral outcome variable.

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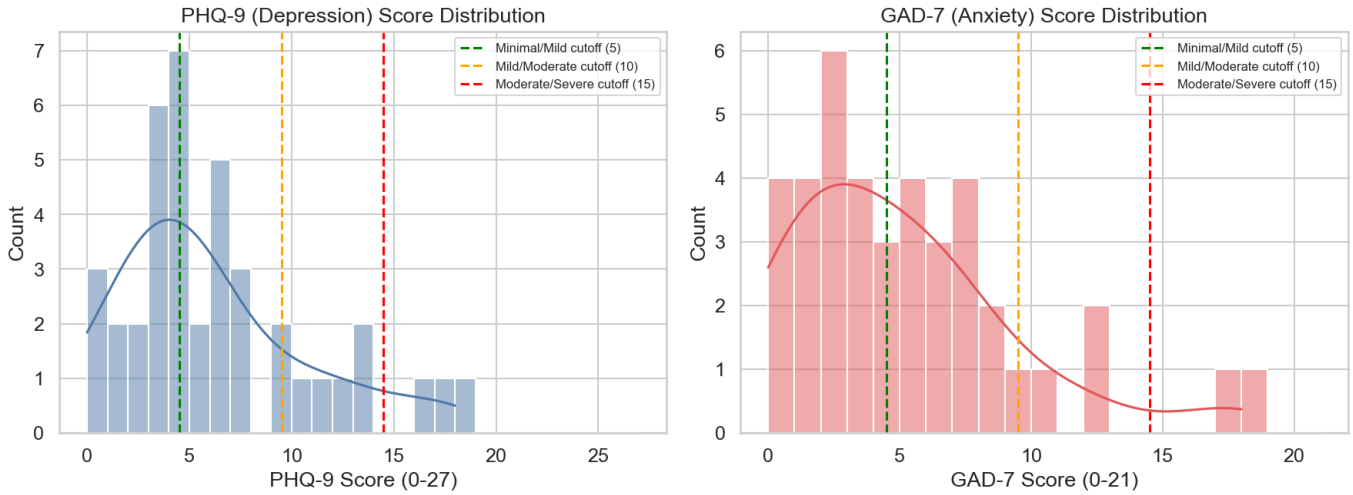
6. Depression & Anxiety Score Distributions

Figure 2: Clinical Score Distributions with Standard Cutoffs



Our sample distribution:

Group	PHQ-9 (Depression)	GAD-7 (Anxiety)
Minimal (0-4)	20 (50%)	21 (52.5%)
Mild (5-9)	12 (30%)	14 (35%)
Moderate-Severe (≥10)	8 (20%)	5 (12.5%)

In the above table each row corresponds to no.of people in the experiment lie in the particular category, which is obtained by comparing the score of participants in PHQ-9 and GAD-7. for example minal (0 - 4) has 20 people whose PHQ-9 score is in the range 0-4 and 21 people with score 0-4 in GAD-7, similarly for other rows.

Compared to the Srivastava, P. & Lahane, R. (2025) paper, where they had 40% moderate-severe depression, our sample has only **20% moderate-severe depression**, reducing group sizes and statistical power.

> Our sample is skewed toward minimal/mild depression — the moderate-severe group (n=8) is small, limiting power to detect effects.

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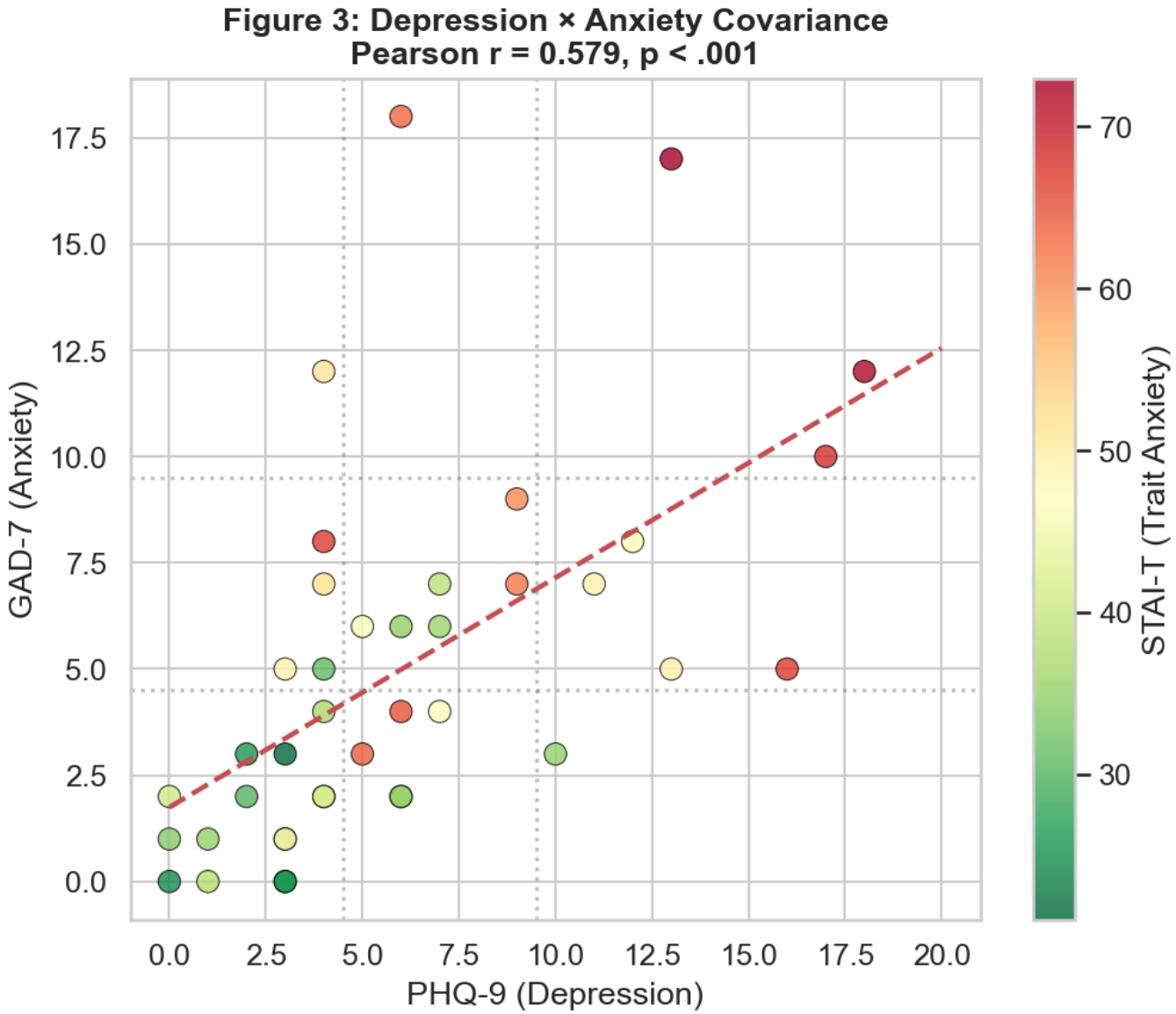
7. Hypothesis, Statistical Tests & Depression–Anxiety Covariance

Hypotheses

H1: Individuals with higher PHQ-9 scores will exhibit lower headtracking scanning speeds during 360° video exploration. **H2**: This effect will be most pronounced for yaw (horizontal scanning) speed

Depression × Anxiety Covariance

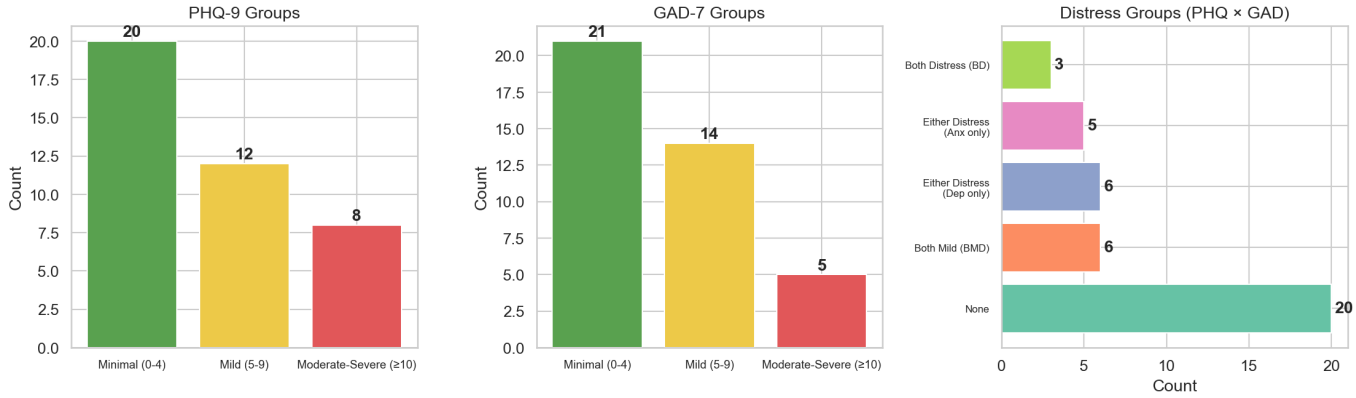
Depression and anxiety are highly comorbid. In our sample:



- **PHQ-9 × GAD-7**: r = .579, p < .001
- **PHQ-9 × STAI-T**: r = .642, p < .001

Following the reference paper's approach, we created **distress groups** based on PHQ-9 × GAD-7 covariance:

Figure 4: Participant Grouping Based on Clinical Cutoffs & Co-occurring Distress



Distress Group	Definition	N
None	Both PHQ-9 and GAD-7 < 5	20
Both Mild (BMD)	Both scores 5–9	6
Either Distress (Dep only)	PHQ ≥ 5, GAD < 5	6
Either Distress (Anx only)	GAD ≥ 5, PHQ < 5	5
Both Distress (BD)	Both ≥ 10	3

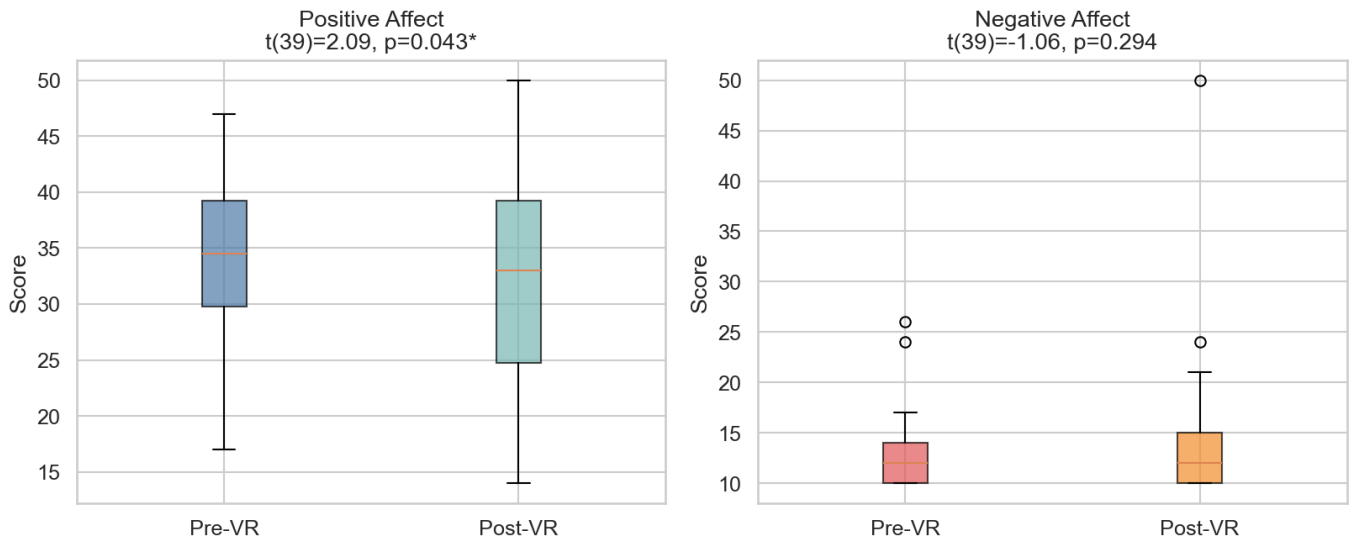
Statistical Tests Used and why we used it

- **Welch's t-tests** (Minimal vs. Moderate-Severe group comparisons): Selected over Student's t-tests because of unequal sample sizes between depression groups (n=20 vs. n=8). The unequal variances between groups are also possible and can be addressed using this statistical approach.
- **Kruskal-Wallis test** (comparing 3 PHQ groups): Selected as it is a non-parametric alternative to one-way ANOVA, as per the reference paper. The data has small and unequal sample sizes, and hence, it is not likely to be normally distributed. The normality assumptions are necessary for performing ANOVA.
- **Pearson correlation** (PHQ-9/GAD-7/STAI-T and headtracking speed): Selected as it is used to examine linear associations between clinical scores and headtracking speed. This approach treats depression and anxiety as continuous variables rather than categorical groups and hence is more appropriate given our sample size.
- **Paired t-tests** (for PANAS pre and post VR): This is appropriate because the data will be the same participants before and after the VR exposure. This will control for individual differences and allow us to assess the impact of the VR exposure on the participants' moods.
- **Use of Cohen's d** (for calculating the effect sizes for each of the tests): This will be used in addition to the p-values because, with such small sample sizes, the effect sizes will give us information on the magnitude of the effects, regardless of their statistical significance, and allow us to compare the results with the reference paper's large effects (d = -1.22 to -1.64).

> Depression and anxiety are strongly correlated (r=.58), necessitating covariance based grouping; Both Distress group has only n=3, limiting distress analysis.

8. Pre-VR and Post-VR Mood (PANAS)

Figure 5: PANAS Mood — Before vs. After VR Experience



Affect	Pre-VR M (SD)	Post-VR M (SD)	t(39)	p
Positive	34.52 (7.33)	32.15 (10.30)	2.09	**.043***
Negative	12.70 (3.52)	13.95 (6.80)	−1.06	.294

Positive affect significantly decreased after VR exposure, while negative affect did not significantly change. The decrease in positive mood may reflect the emotionally heavy content of certain videos (e.g., The Nun Horror).

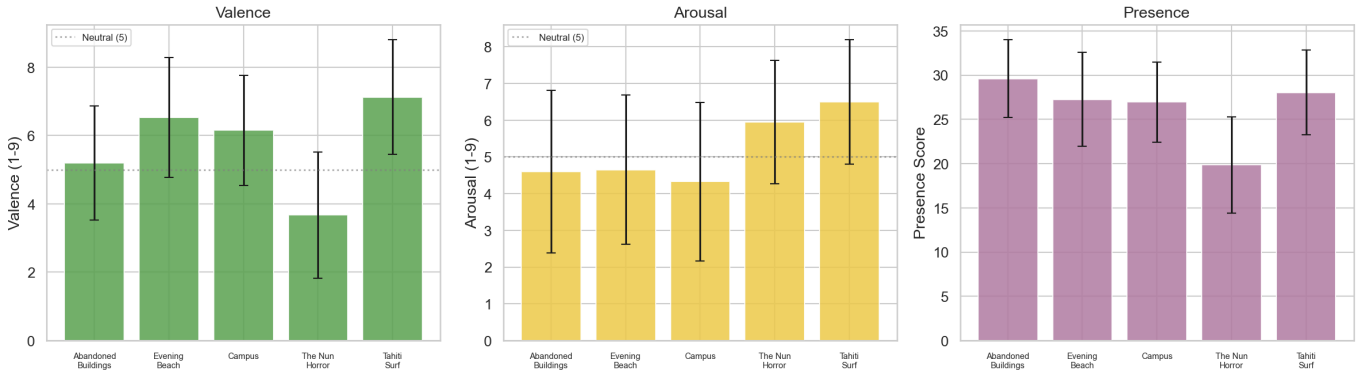
> VR exposure reduced positive mood significantly — the experience was not neutral, which is important context for interpreting headtracking results.

9. Why Different Videos? — Valence, Arousal & Presence

The 5 videos were selected to span **different emotional conditions**, ensuring headtracking behavior is measured across diverse stimuli:

Video	Expected Emotion	Valence M	Arousal M
V1: Abandoned Buildings	Eerie/Unsettling	5.20	4.60
V2: Evening at Beach	Pleasant/Calm	6.53	4.65
V3: Campus	Familiar/Neutral	6.15	4.33
V4: The Nun Horror	Fearful/Intense	3.67	5.95
V5: Tahiti Surf	Pleasant/Exciting	7.12	6.50

Figure 6: Why Different Videos? — Valence, Arousal & Presence Across Emotional Conditions



Key observations:

- Videos successfully span different emotional conditions — from unpleasant/intense (Horror) to pleasant/exciting (Surf)
- **V4 (Horror) had notably lower presence scores** (M = 19.85 vs. ~27–30 for others), suggesting fear may break immersion
- Using multiple video types tests whether depression effects are **consistent across stimuli** or **stimulus-specific**

> The 5 videos span different emotional regions, enabling us to test whether depression reduces scanning across ALL stimuli (supporting emotional context insensitivity).

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10. Visualizations & Results

10.1 Headtracking Speed by Depression Group (Clinical Cutoffs)

Figure 7: Headtracking Speed by Video × PHQ-9 Depression Group

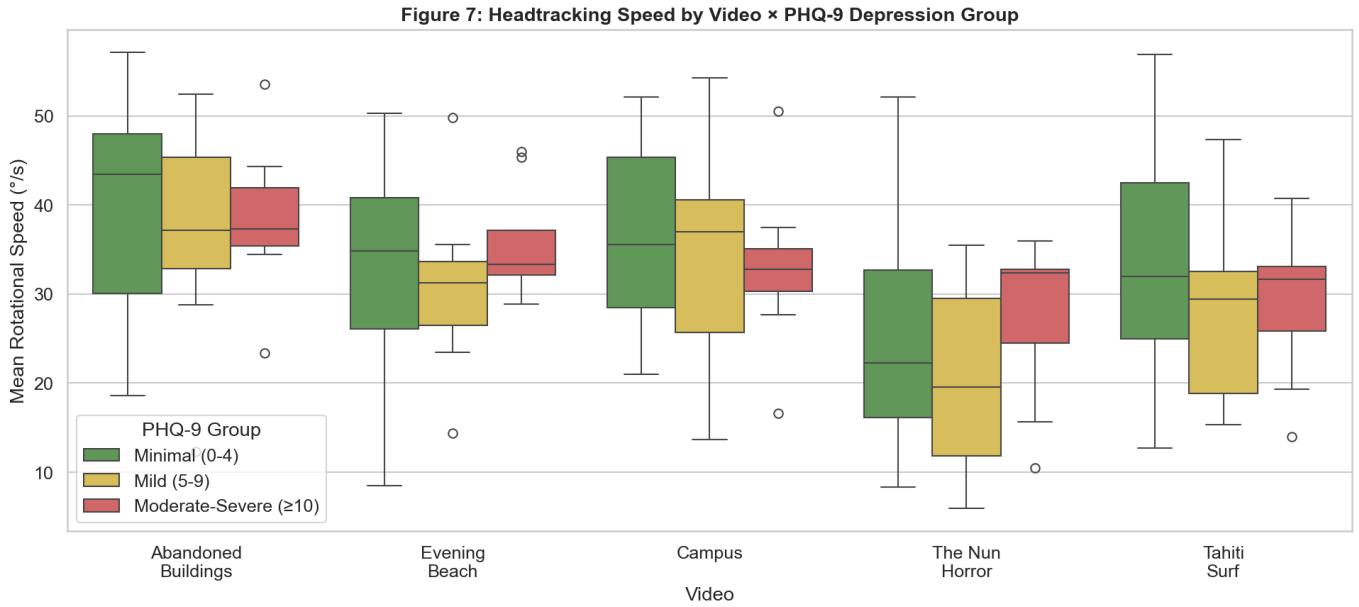


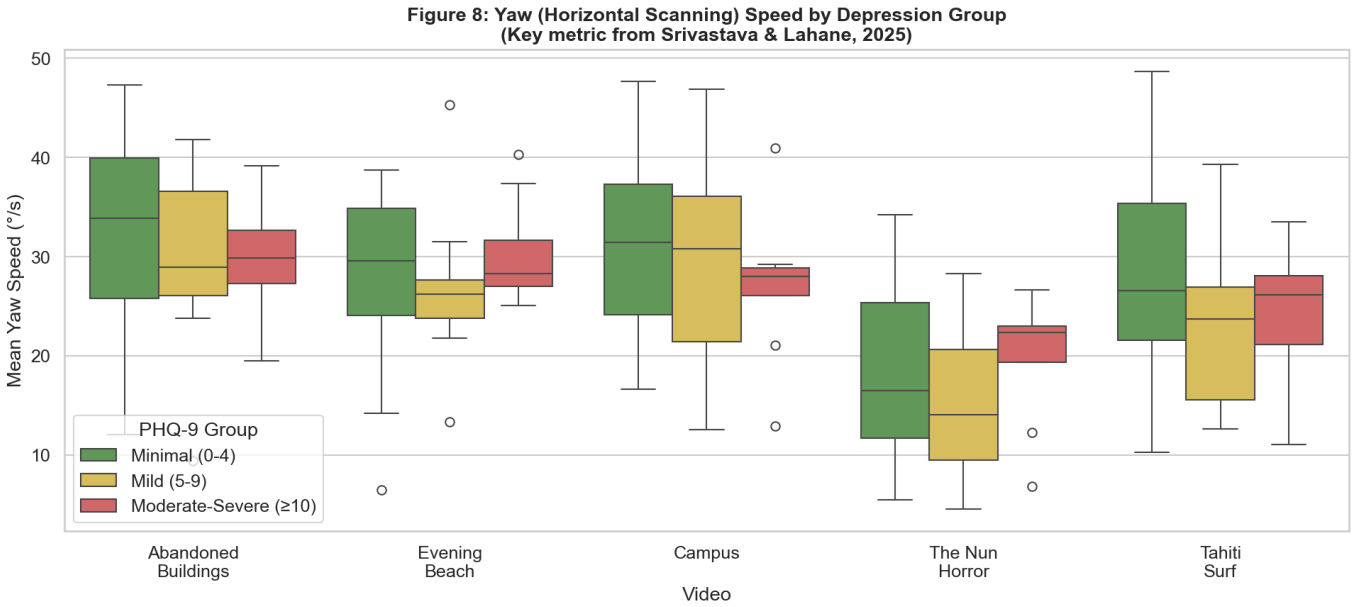
Table: Welch's t-tests — Minimal vs. Moderate-Severe Depression

Video	Mod-Severe M (SD)	Minimal M (SD)	t	p	Cohen's d
Abandoned Buildings	38.39 (8.65)	40.34 (11.80)	−0.48	.635	−0.19

Video	Mod-Severe M (SD)	Minimal M (SD)	t	p	Cohen's d
Evening at the Beach	35.55 (6.50)	32.88 (10.34)	0.82	.422	0.31
Campus	32.89 (9.47)	36.56 (10.25)	−0.90	.382	−0.37
The Nun Horror	27.49 (9.32)	25.14 (12.04)	0.55	.589	0.22
Tahiti Surf	29.09 (8.62)	33.78 (12.52)	−1.13	.271	−0.44

All p-values are non-significant. The **Tahiti Surf** and **Campus** videos show the largest (still non-significant) effects in the expected direction (d = −0.44, −0.37).

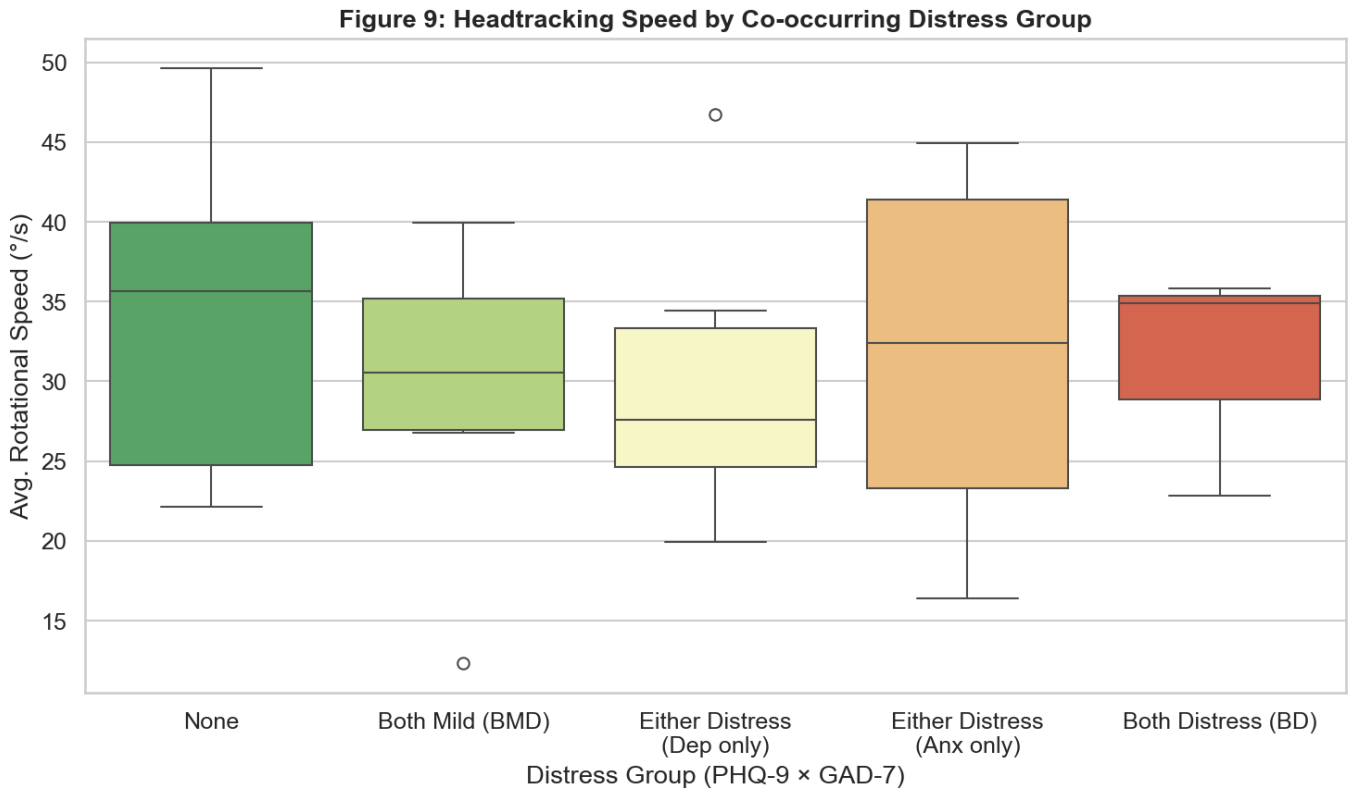
10.2 Yaw (Horizontal Scanning) Speed — Key Metric



Yaw speed — the primary metric in Srivastava & Lahane (2025) — similarly shows **non-significant** differences (all p > .24).

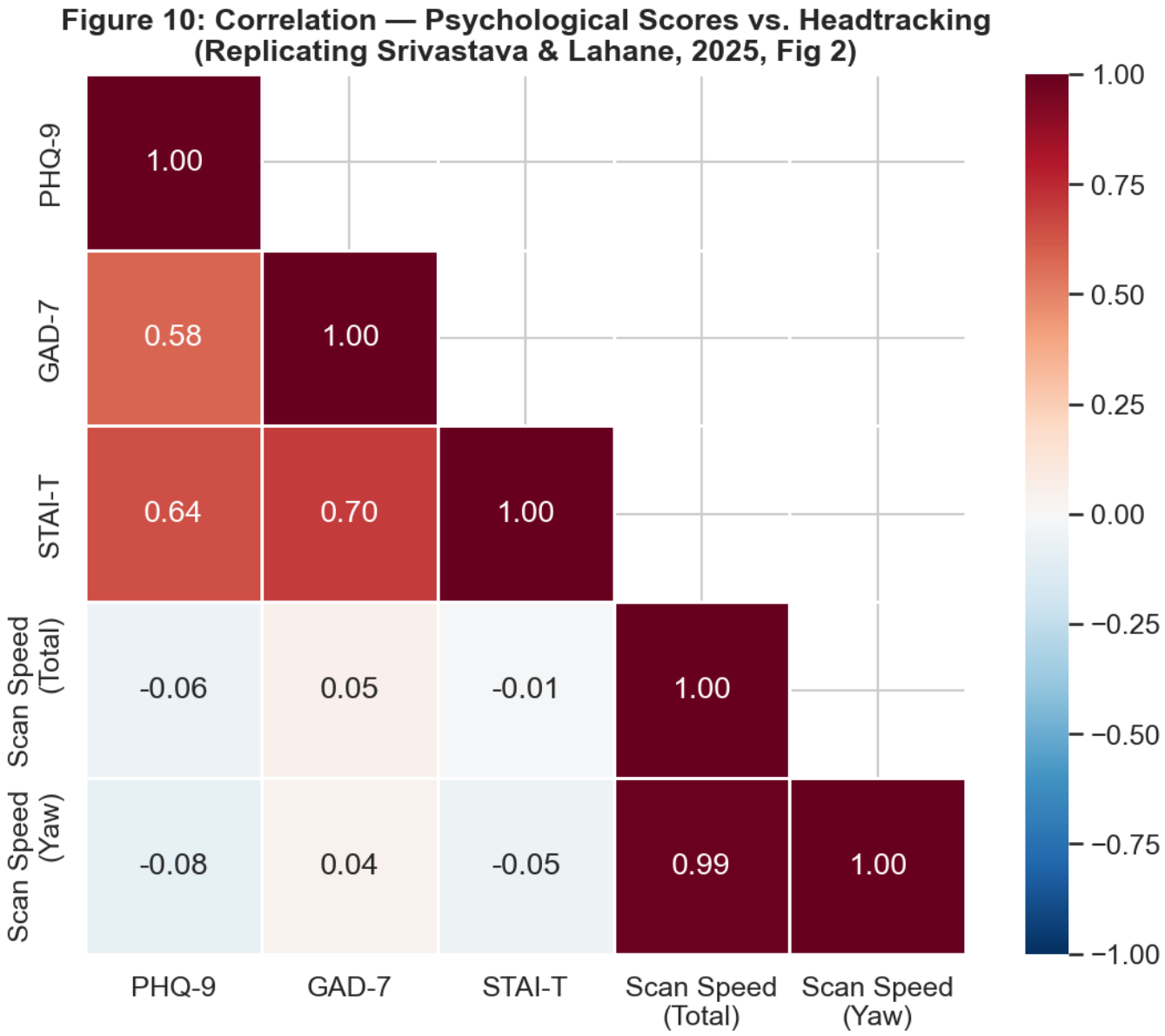
Kruskal-Wallis (non-parametric, replicating paper): Total Speed H = 0.90, p = .639; Yaw Speed H = 1.11, p = .573.

10.3 Headtracking by Distress Group (Covariance-Based)



The distress grouping shows some visual trends — "Both Distress" participants tend toward lower speeds — but group sizes are too small (BD: n = 3) for meaningful inference.

10.4 Correlation Matrix



Pair	r	p
PHQ-9 × GAD-7	.579	< .001
PHQ-9 × STAI-T	.642	< .001
PHQ-9 × Scan Speed	−.060	.712
PHQ-9 × Yaw Speed	−.081	.617
GAD-7 × Scan Speed	.048	.768

Correlations between psychological scores and headtracking speed are **weak and non-significant**, contrasting with the reference paper's moderate negative correlations.

> No statistically significant headtracking differences between depression groups — likely due to our smaller sample (N=40, only 8 moderate-severe) compared to the reference paper (N=50, 20 moderate-severe).

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11. Conclusions

1. **Direction partially consistent**: The moderate-severe depression group had lower scanning speed for Campus and Tahiti Surf videos, which is consistent with the direction taken in the reference paper; however, the results are not significant.
2. **Self reported emotions worked as expected**: Videos induced the expected emotions, which were validated through valence and arousal ratings; VR exposure also had the expected significant effect on reducing positive emotions.
3. **Replication not achieved**: In contrast to Srivastava & Lahane (2025), who reported significant differences in scanning speed between groups ($p < .001$), our results do not show significant differences.
4. **Primary limitation, Small sample size**: With only **n = 8** participants for the moderate-severe depression group (compared to n = 20 for the reference paper), we cannot achieve significant results. Moreover, our participants only had 20% moderate-severe depression, compared to 40% for the reference study.
5. **Depression-anxiety covariance**: Since PHQ-9 and GAD-7 are highly correlated ($r = .579$), depression's effects must be isolated from anxiety's effects; however, our small n does not allow for ANCOVA analysis.

> The study is underpowered to detect the moderate effects found in the reference paper; larger/more clinically diverse samples are needed to replicate the finding.

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12. Plan for Report 2

1. **ANCOVA**: Include GAD-7 and STAI-T as covariates to analyze the impact of depression on scanning speeds, similar to the reference paper.
2. **Regression/GLM**: Include PHQ-9 as a continuous variable to increase our ability to detect differences with our small sample size.
3. **Mixed-effects models**: Include video type as a within-subjects variable to determine whether a depression x video type interaction is present.
4. **Advanced head tracking features**: Examine standard deviations of yaw (SDY) and temporal dynamics (speed trajectories throughout a video's duration), as in the reference paper.
5. **Power analysis**: Determine the required sample size to detect the reference paper's effect sizes ($d = 1.2 - 1.6$) and discuss our results in this context.
6. **Normality checks**: Apply Shapiro-Wilk tests, as in the reference paper, to determine which parametric or non-parametric tests to perform on each variable.

Contributions

All members have **contributed equally on all tasks** and report writing, with each member taking special responsibility for the following:

Member	Contributions
Bharath Gajawada	Data preprocessing, demographic analysis, clinical score distributions (PHQ-9, GAD-7), depression–anxiety covariance analysis and distress group creation

Member	Contributions
Vishnu Sai Reddy	Headtracking data analysis (scanning speed, yaw speed by depression groups), Welch's t-tests, Kruskal-Wallis tests, correlation analysis, generating all statistical figures
Pooja Rani	Literature review, PANAS pre/post analysis, valence/arousal/presence analysis across videos, interpretation of results

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